

Anomaly Detection Results

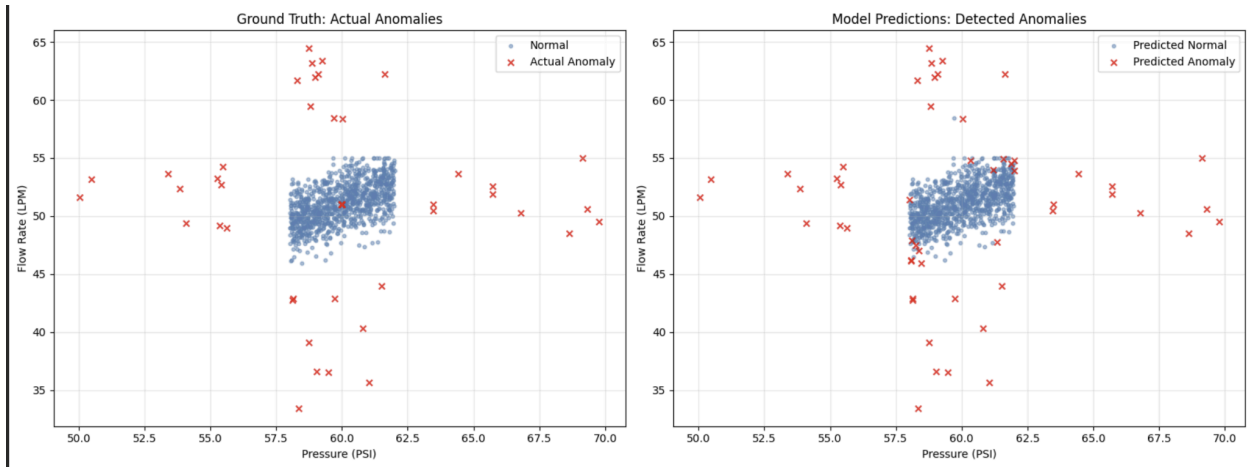
Anomaly Detection Results

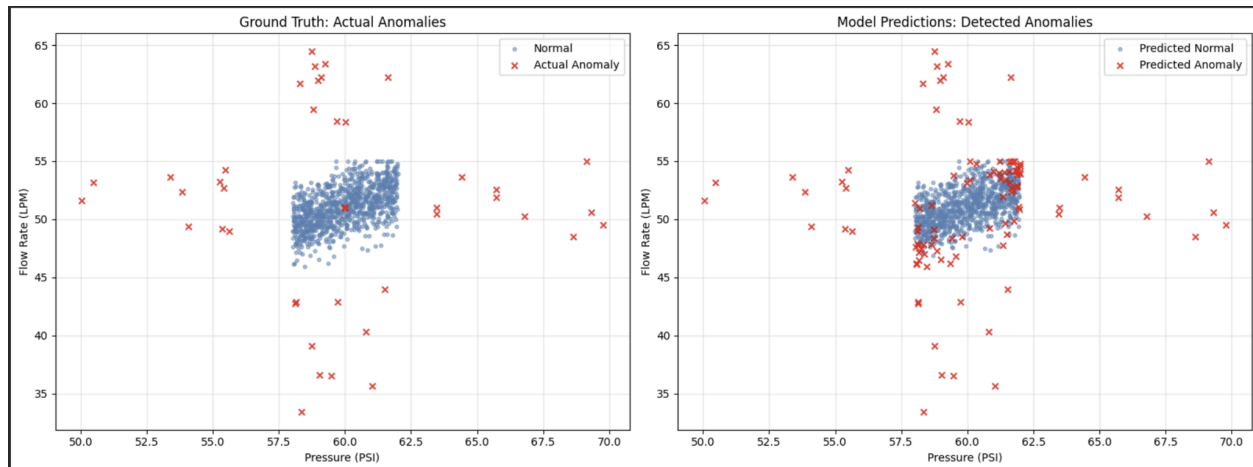
Baseline Results

After running the notebook, copy the baseline results here:

Metric	Value
Total data points	1050
Anomalies injected	50
Anomalies detected by the model	53
True positives (correctly flagged)	39
False positives (normal data flagged)	14
False negatives (anomalies missed)	11
Precision	0.736
Recall	0.780
F1 Score	0.757

Scatter Plot Observations





Parameter Experiment

You changed **contamination** from **0.05** to **0.1**. Record the comparison below:

Metric	Default (0.05)	Changed (0.1)
Anomalies detected	53	105
True positives	39	40
False positives	14	65
Precision	0.736	0.381
Recall	0.780	0.800
F1 Score	0.757	0.516

Write one sentence explaining what changed when you increased contamination:

The model expects more anomalies, thereby reducing the threshold of anomalies, and this leads to high Recall but low Precision

Hyperparameter Experiments

Hyperparameter Experiments

Contamination Sweep

Contamination	Anomalies Detected	Precision	Recall	F1 Score
0.01	11	1.000	0.220	0.361
0.03	32	1.000	0.640	0.780
0.05	53	0.736	0.780	0.757
0.1	105	0.381	0.800	0.516
0.2	210	0.190	0.800	0.308

Which contamination value gives the best F1 score?

A contamination of 0.03

Tree Count Experiment

n_estimators	Precision	Recall	F1 Score	Training Time (sec)
50	0.698	0.740	0.718	0.201s
100	0.736	0.780	0.757	0.484s
200	0.717	0.760	0.738	0.880s
500	0.717	0.760	0.738	2.743s

At what point do more trees stop improving the results?

After a hundred trees.

Does training time increase linearly with the number of trees?

Yes, the higher the number of trees, the greater the time it takes to train them.

Sample Size Experiment

max_samples	Precision	Recall	F1 Score
auto	0.736	0.780	0.757
64	0.585	0.620	0.602
128	0.698	0.740	0.718
256	0.736	0.780	0.757

Which sample size gives the best F1 score?

Auto which is (256, n_samples)

Best Configuration

Metric	Default	Best	Change
precision	0.736	0.736	+ 0.000
recall	0.780	0.780	+ 0.000
f1	0.757	0.757	+ 0.000

Write 2-3 sentences analyzing the difference:

The model chose the best configuration

Model Comparison

Model Comparison

LOF Default Results

```
=====
LOF (n_neighbors=20) Results
=====
True Positives (correctly flagged): 40
False Positives (normal flagged): 13
False Negatives (anomalies missed): 10
True Negatives (correctly passed): 987
---
Precision: 0.755
Recall: 0.800
F1 Score: 0.777
=====
Training time: 0.055s
```

LOF Tuning (n_neighbors)

n_neighbors	Precision	Recall	F1	Time
10	0.755	0.800	0.777	0.033s
20	0.755	0.800	0.777	0.037s
50	0.755	0.800	0.777	0.117s

Which n_neighbors value gives the best F1 score?

10,20 and 50

Head-to-Head Comparison

```
=====
HEAD-TO-HEAD: Isolation Forest vs. Local Outlier Factor
=====
```

Metric	IF (Best)	LOF (Best)
Precision	0.736	0.755
Recall	0.780	0.800
F1 Score	0.757	0.777

Training Time		0.344s		0.033s
Can score new data one at a time?		Yes		No
Needs full dataset present?		No		Yes

Overall winner by F1: Local Outlier Factor

Note: Even if LOF has slightly better F1, Isolation Forest is the better choice for real-time IoT because it can score new messages one at a time as they arrive from the MQTT broker.

Your Recommendation

Which model should the Grand Marina deploy for real-time anomaly detection on their water system, and why? Write 3-5 sentences.

LOF achieved higher F1 but is unsuitable for real-time IoT deployment. Isolation Forest was selected as the production model due to its ability to score individual messages as they arrive without retraining.

Optional: One-Class SVM Results

```
=====
One-Class SVM Results
=====
True Positives (correctly flagged): 33
False Positives (normal flagged): 21
False Negatives (anomalies missed): 17
True Negatives (correctly passed): 979
---
Precision: 0.611
Recall: 0.660
F1 Score: 0.635
=====
Training time: 0.021s
```

```
=====
THREE-WAY COMPARISON
=====
```

Metric		IF		LOF		SVM

Precision		0.736		0.755		0.611
Recall		0.780		0.800		0.660
F1 Score		0.757		0.777		0.635

Training Time		0.344s		0.033s		0.021s
Real-time scoring?		Yes		No		Yes
Needs scaling?		No		No		Yes